

HTTP vs HTTPS and HTTP Request Methods

Introduction

Whenever we open a website, our browser communicates with a web server. This communication happens through a protocol called HTTP or HTTPS. Although both are used to transfer information, HTTPS is much safer because it protects our data. Knowing the difference between them is important for anyone learning web development.

HTTP vs HTTPS

HTTP (HyperText Transfer Protocol) is the basic protocol used to load web pages. It is fast but does not encrypt the data being shared between the browser and the server. Because of this, sensitive information like passwords can be exposed.

HTTPS (HyperText Transfer Protocol Secure) is the secure version of HTTP. It uses SSL/TLS encryption to keep data safe while it travels over the internet. Most modern websites, especially banking and shopping sites, use HTTPS to protect users.

Difference Between HTTP and HTTPS

1. HTTP is not secure, while HTTPS is secure.
2. HTTP does not encrypt data; HTTPS encrypts all data.
3. HTTP uses Port 80, while HTTPS uses Port 443.
4. HTTPS requires an SSL/TLS certificate; HTTP does not.
5. HTTP is more vulnerable to attacks.
6. HTTPS protects user privacy.
7. Browsers show a padlock icon for HTTPS websites.
8. HTTPS is recommended for online payments and logins.
9. Search engines prefer HTTPS websites.
10. HTTPS builds more trust among users.

HTTP Request Methods

HTTP request methods tell the server what action the client wants to perform.

GET

Used to retrieve data from the server.

Example: Viewing a webpage or fetching user details.

POST

Used to send new data to the server.

Example: Registering a new account or submitting a contact form.

PUT

Used to replace or update an existing resource.

Example: Updating all details of a user profile.

PATCH

Used to update only specific information.

Example: Changing only a user's phone number.

DELETE

Used to remove data from the server.

Example: Deleting a user account or a product.

HEAD

Retrieves only the response headers without the actual content.

OPTIONS

Shows the HTTP methods supported by a server.

CONNECT

Creates a secure connection through a proxy server.

TRACE

Helps developers test and debug HTTP requests.

Conclusion

HTTP and HTTPS are essential for communication on the internet. While HTTP is suitable for basic communication, HTTPS is the preferred choice because it keeps user data safe through encryption. Along with understanding HTTP and HTTPS, learning request methods like GET, POST, PUT, PATCH, and DELETE helps developers build secure and efficient web applications.